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Dallenbach

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(45) **Date of Patent:** ***Aug. 26, 2025**

(54) **ADJUSTABLE FIREARM STEADY REST**
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This patent is subject to a terminal disclaimer.

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Related U.S. Application Data

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F41A 23/56 (2006.01)

(52) **U.S. Cl.**
CPC **F41A 23/16** (2013.01); **F41A 23/56** (2013.01)

(58) **Field of Classification Search**
CPC **F41A 23/16**
See application file for complete search history.

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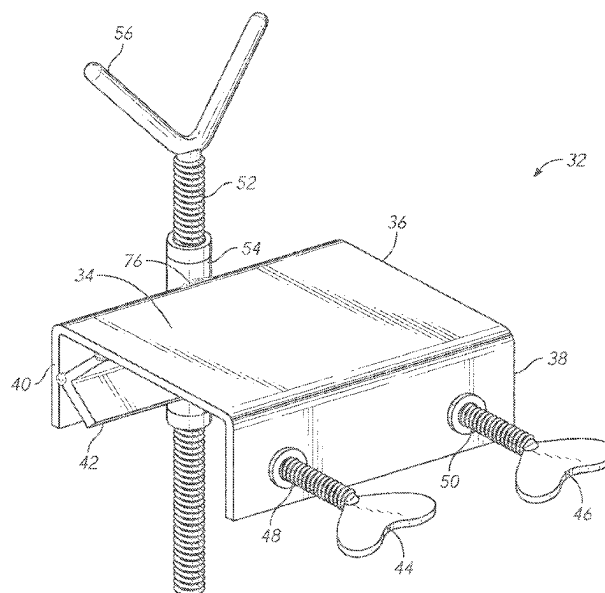
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(57) **ABSTRACT**

An adjustable shooting steady rest configured to clamp to a wide variety of structures. The steady rest includes a chassis with one or more clamps opposing a clamping bracket. The user places the chassis over a structure such as a safety bar or windowsill. The user then tightens the clamp(s) to lock the chassis in place. The chassis includes a rod receiver. A rod is adjustably attached to the rod receiver. A firearm support is provided on the upper portion of the rod.

8 Claims, 19 Drawing Sheets



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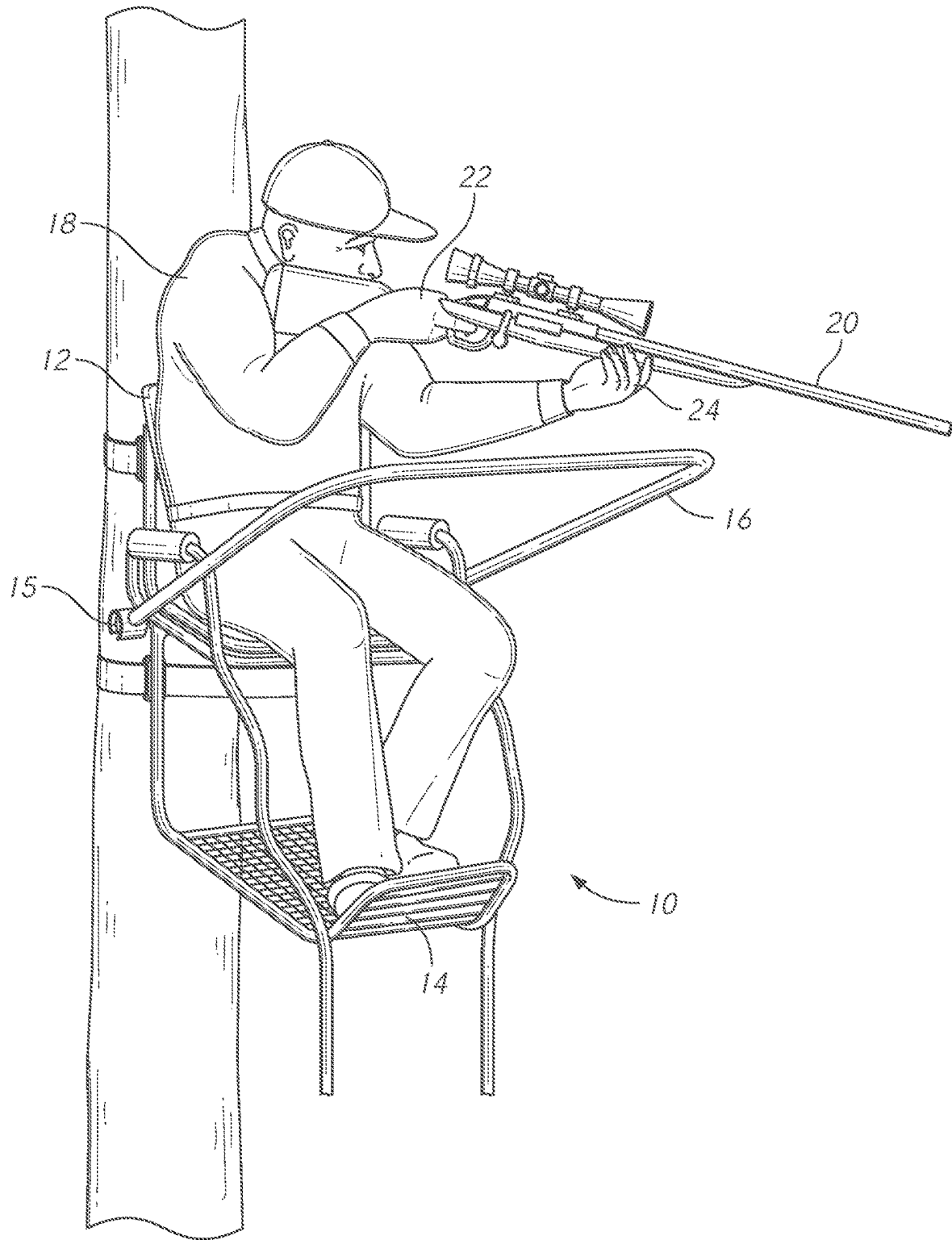


FIG. 1
(Prior Art)

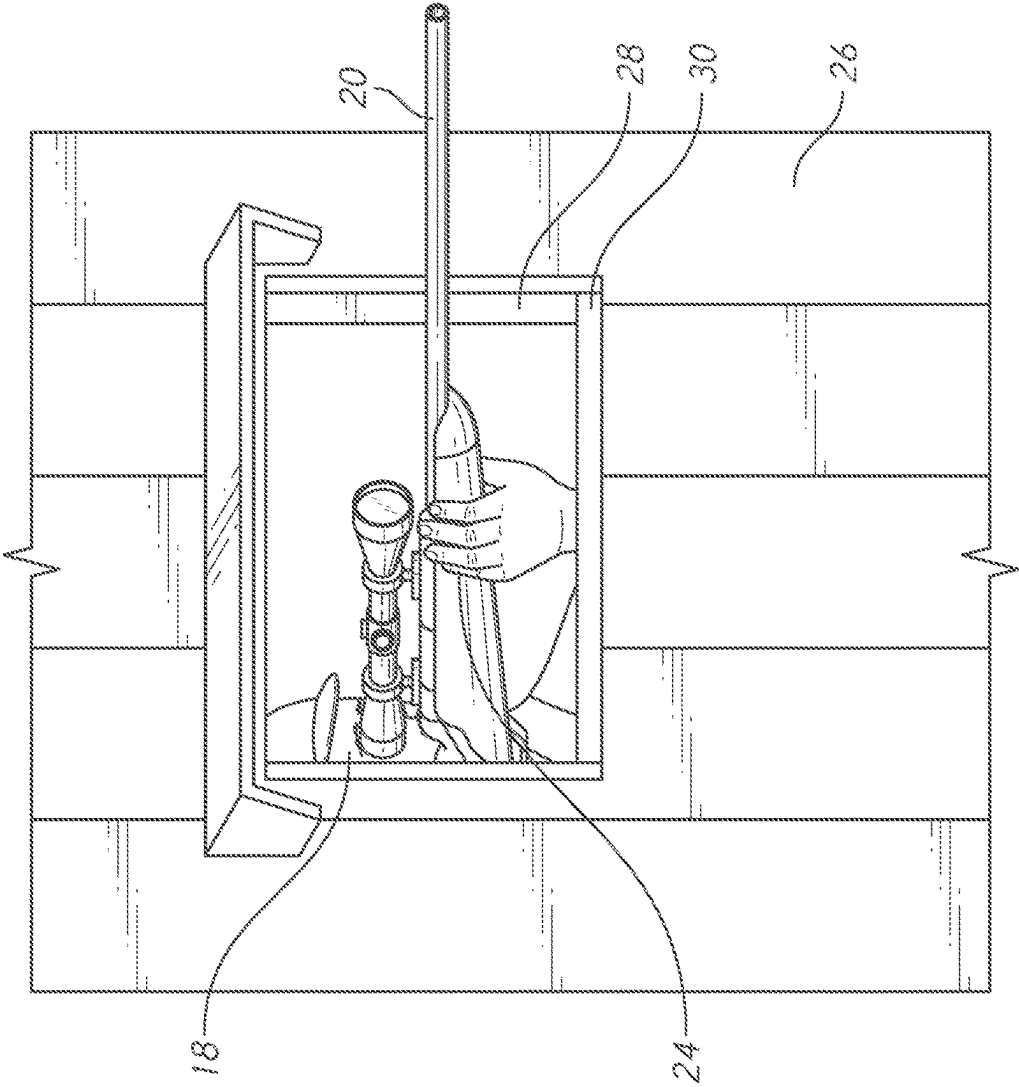


FIG. 2
(Prior Art)

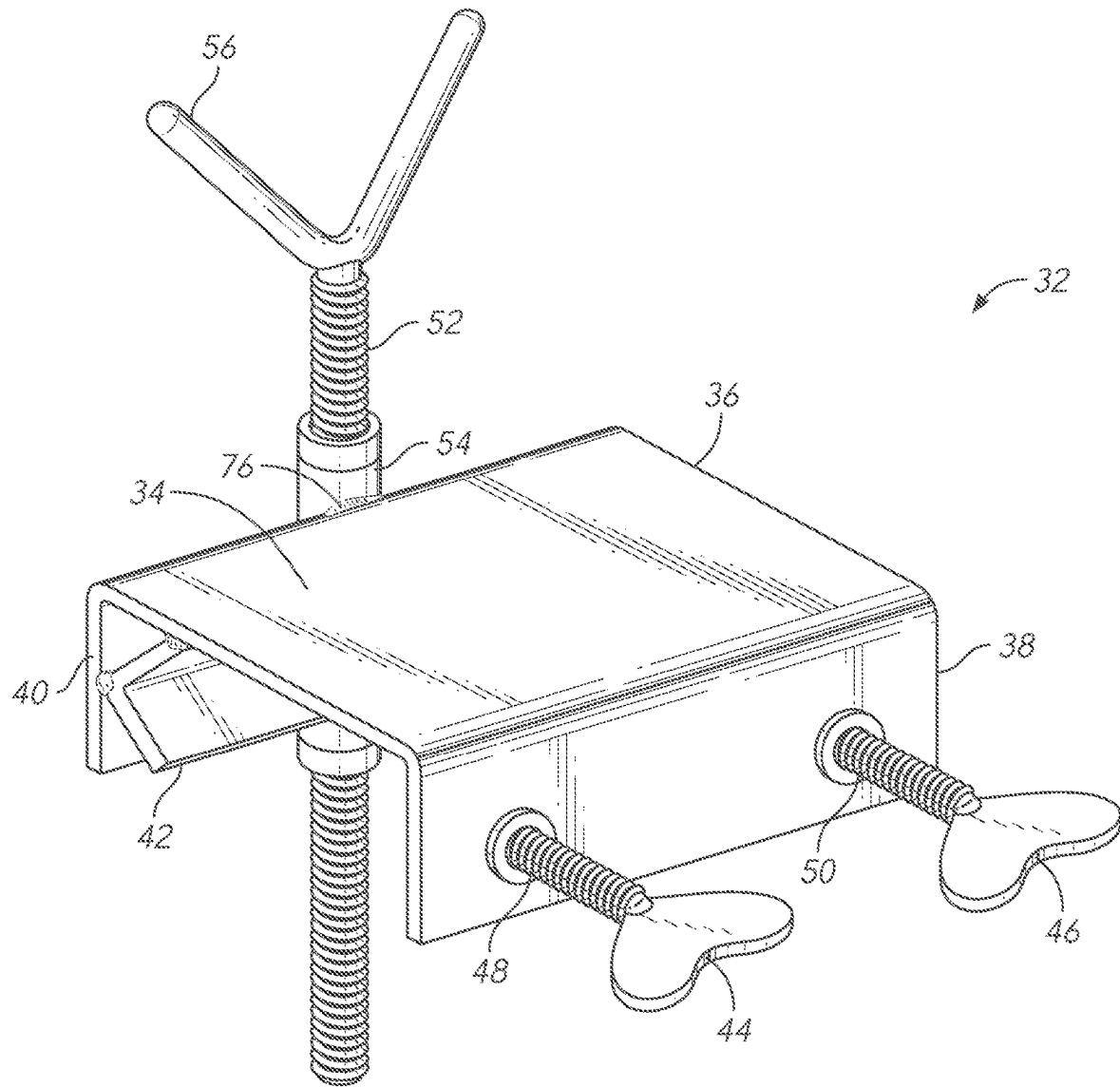


FIG. 3

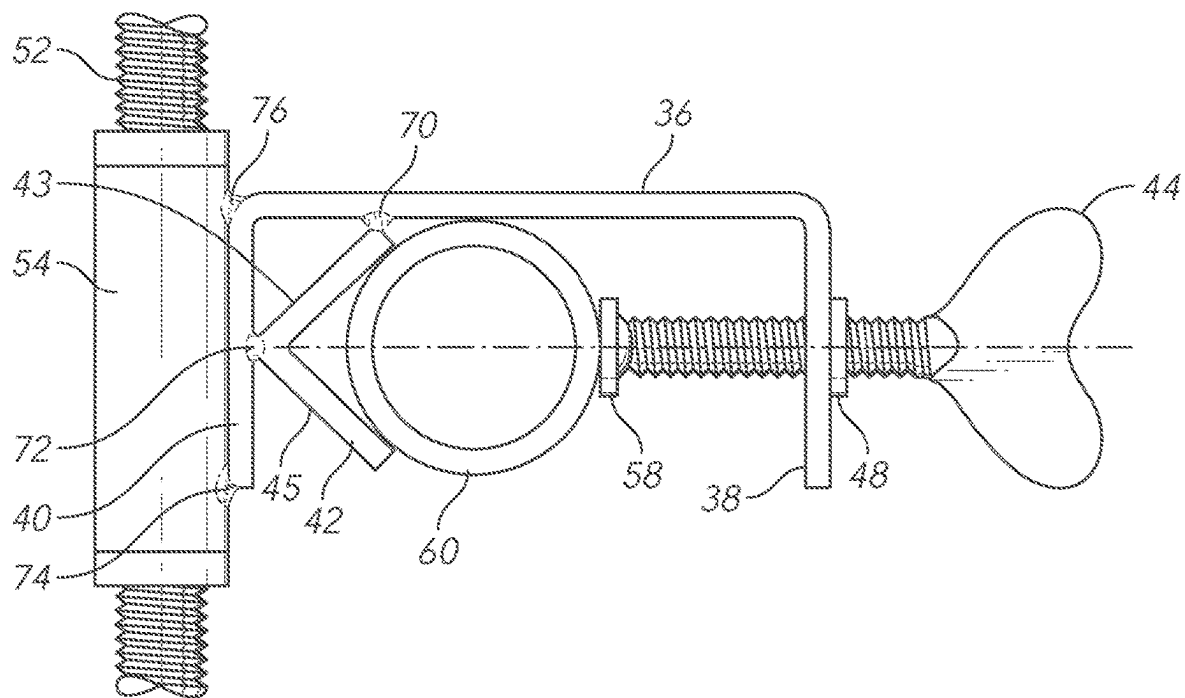


FIG. 4

FIG. 5

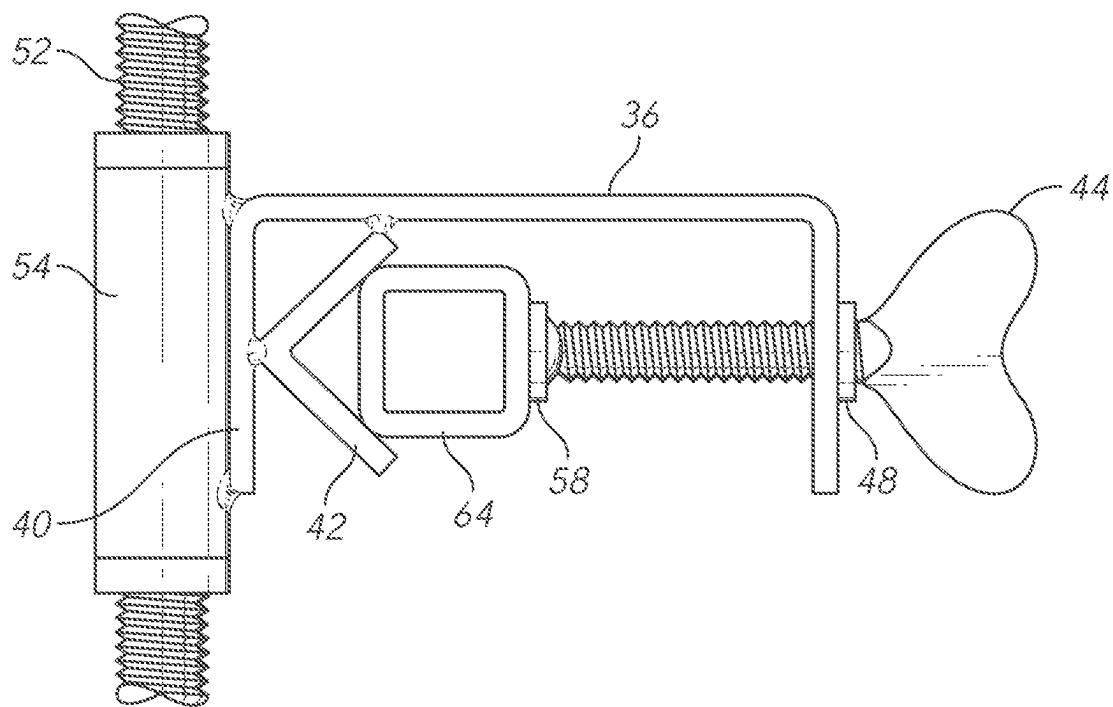


FIG. 6

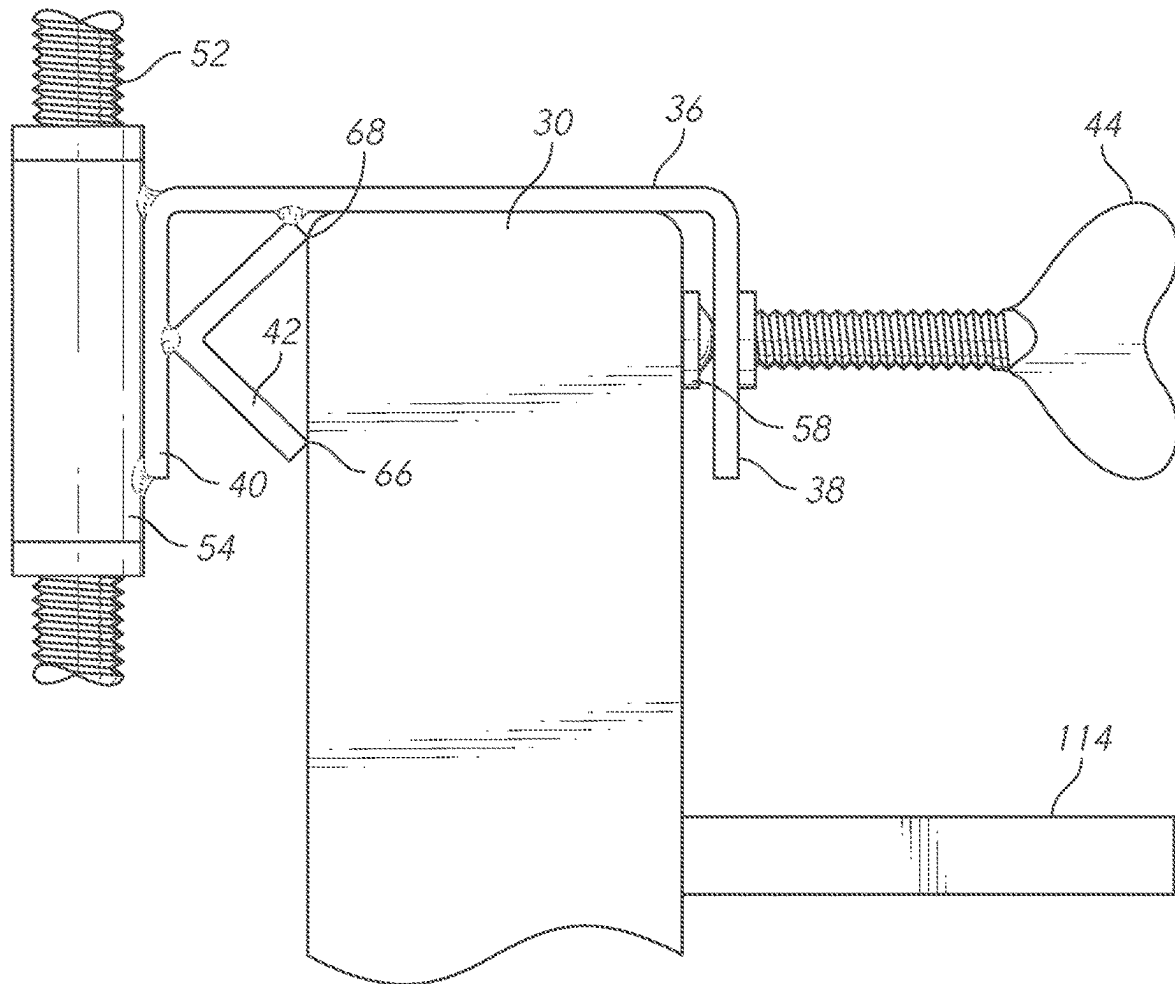


FIG. 7

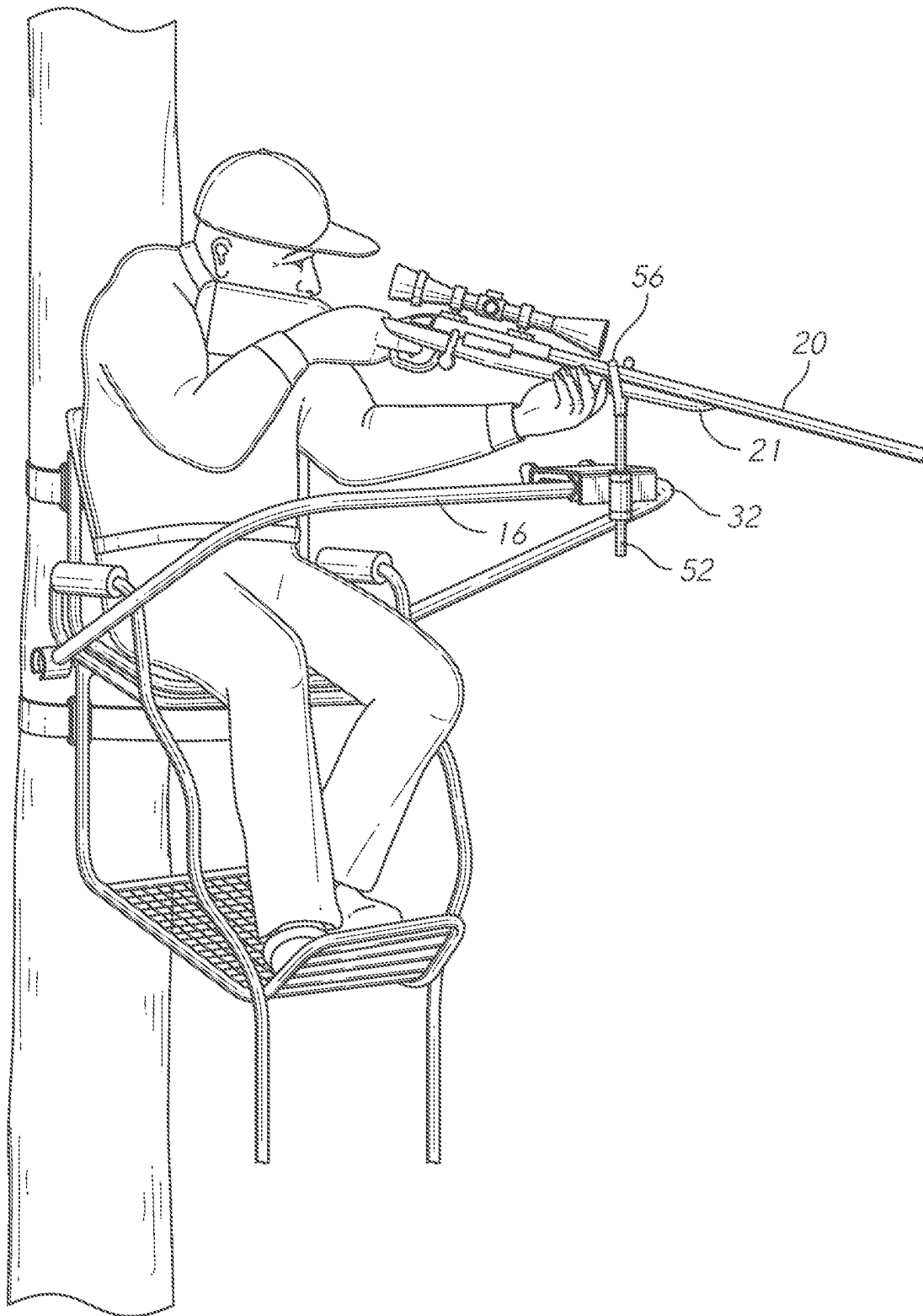


FIG. 8

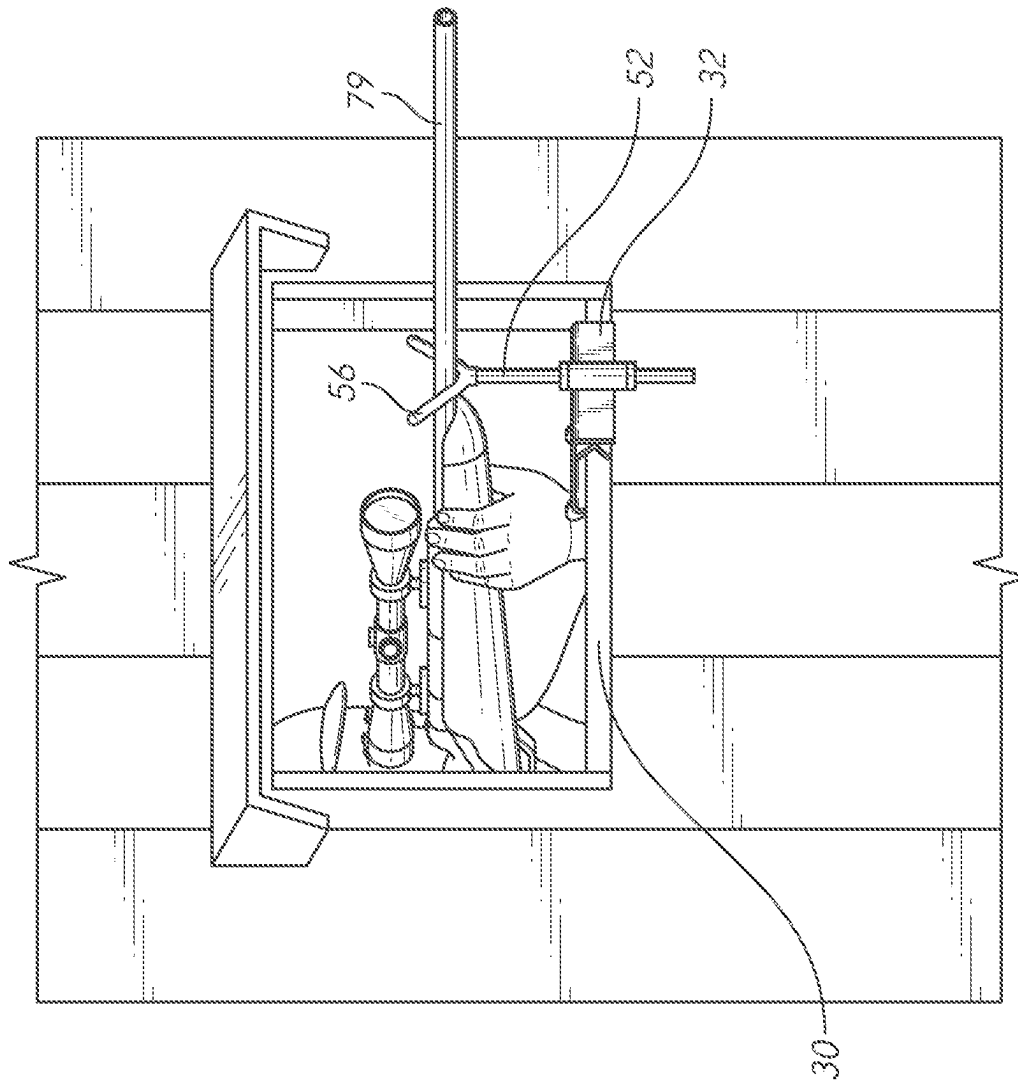


FIG. 9

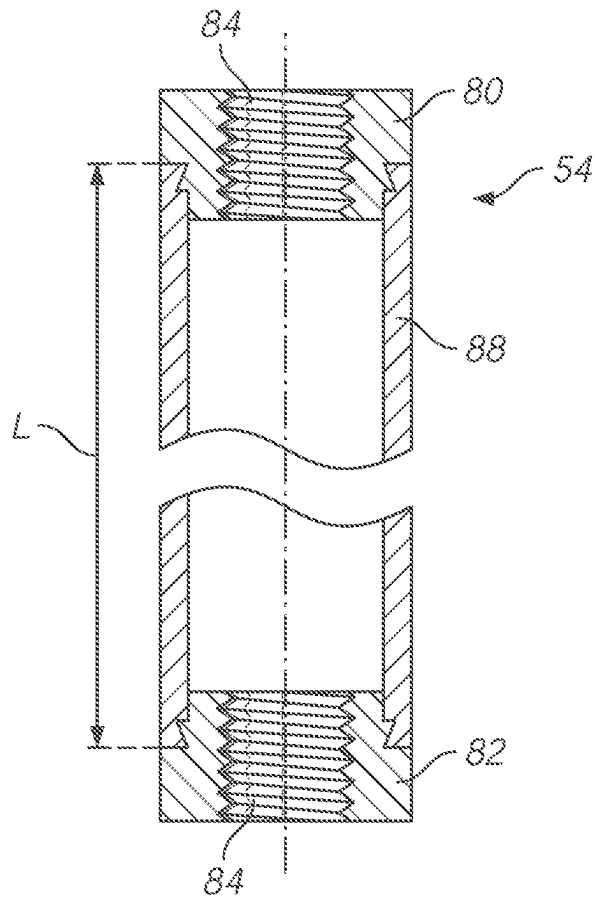


FIG. 10

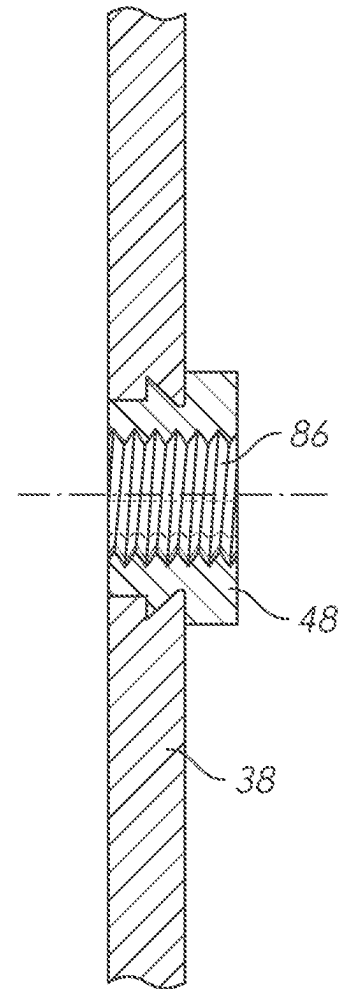


FIG. 11

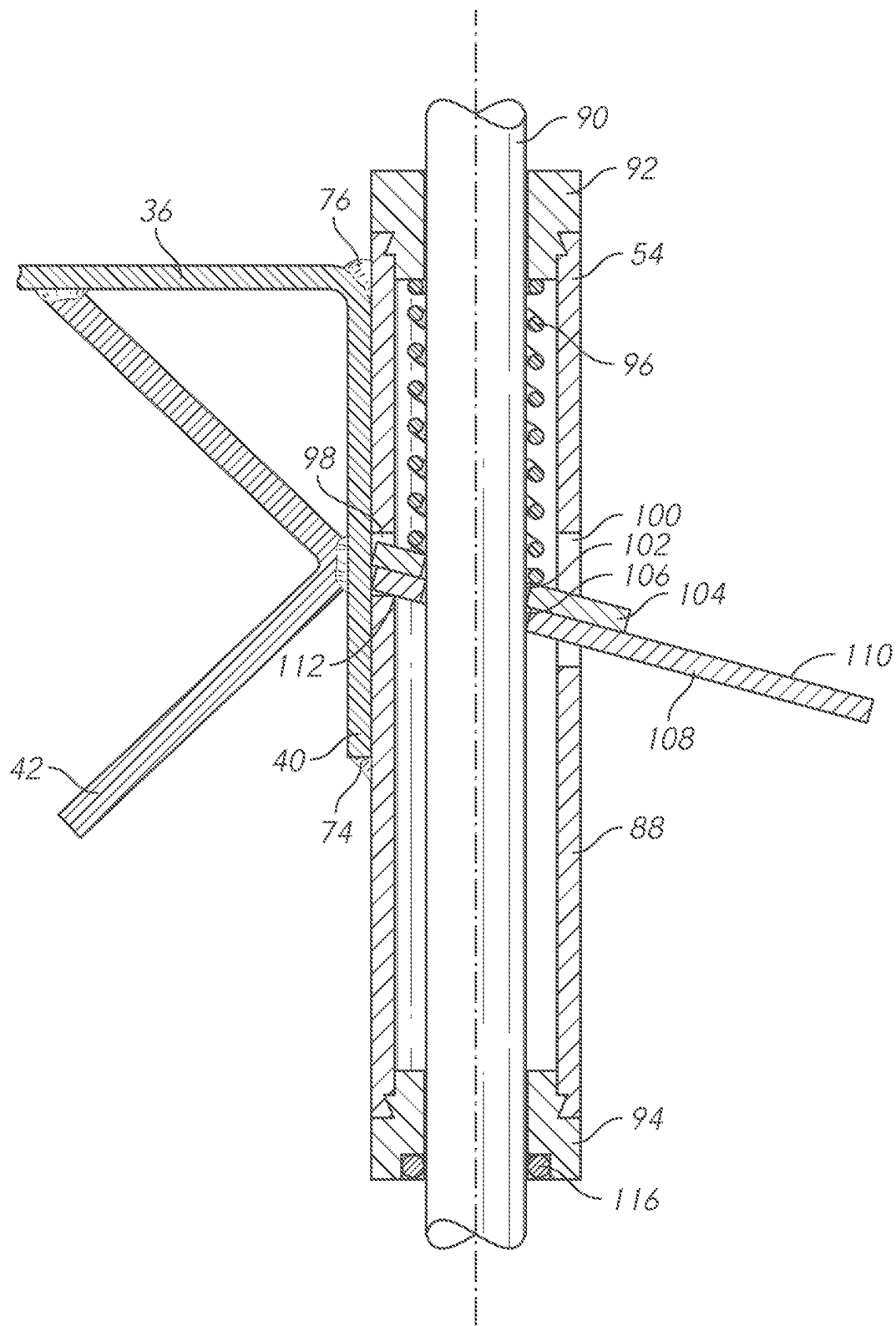


FIG. 12

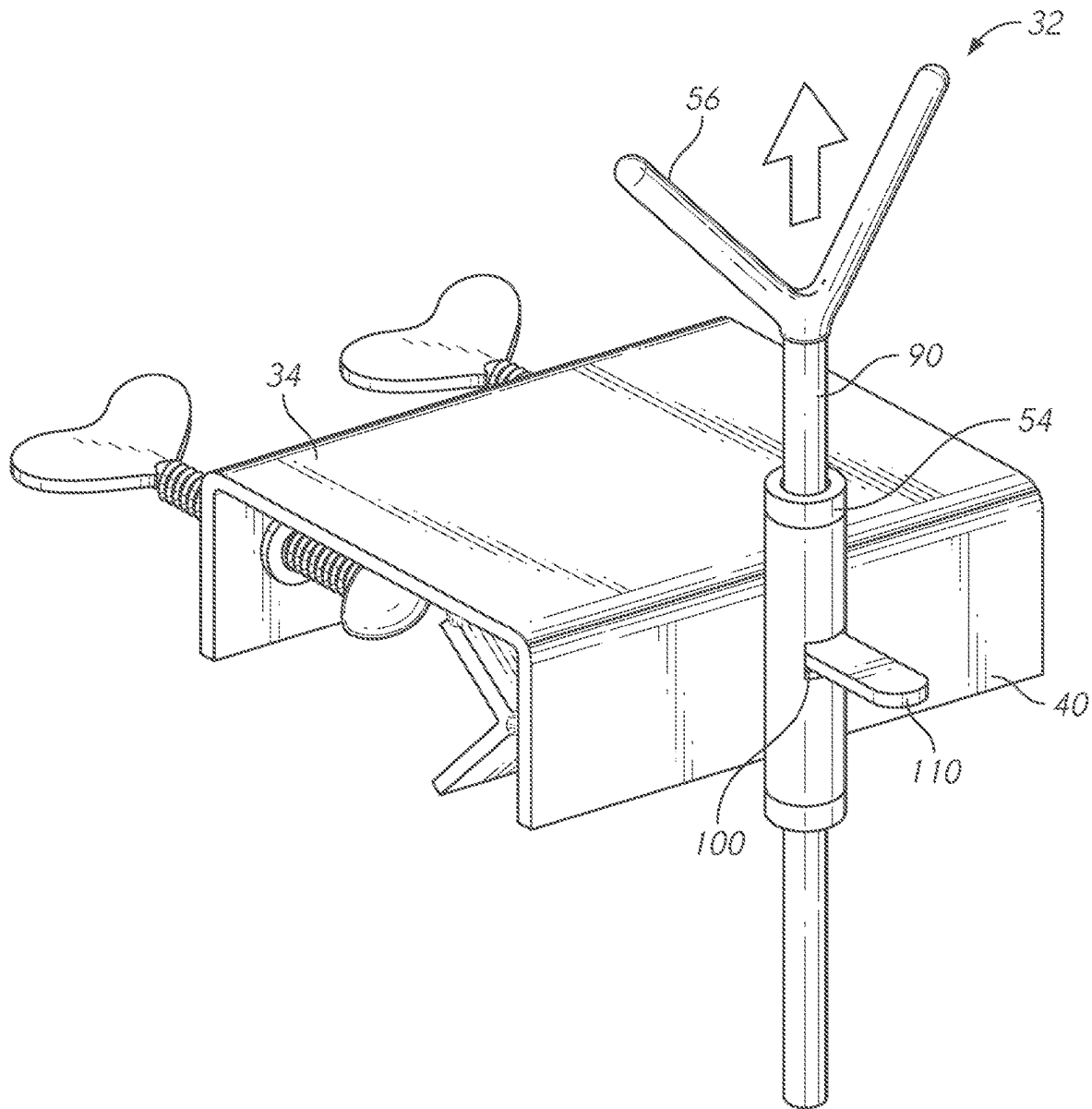


FIG. 13

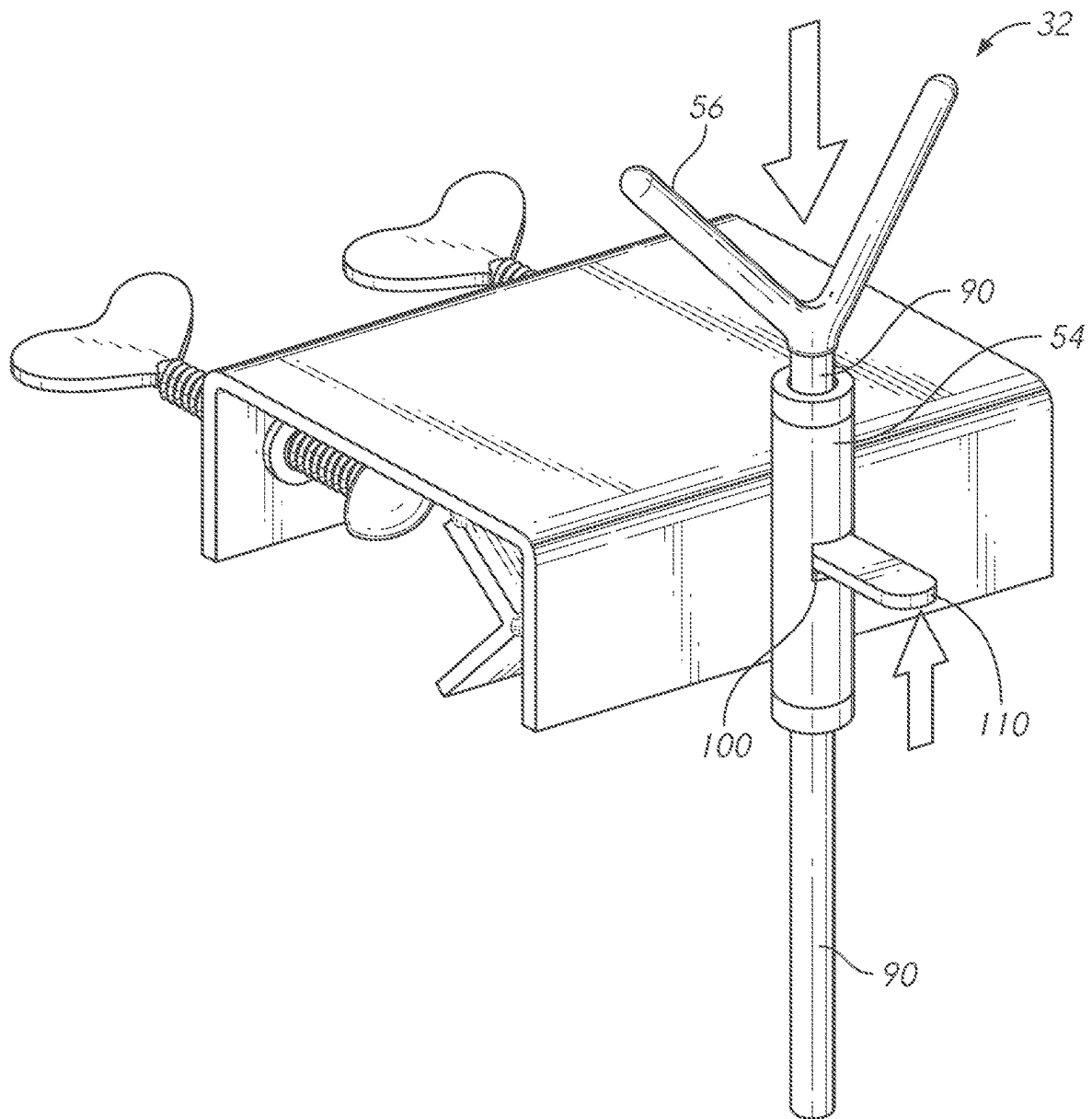


FIG. 14

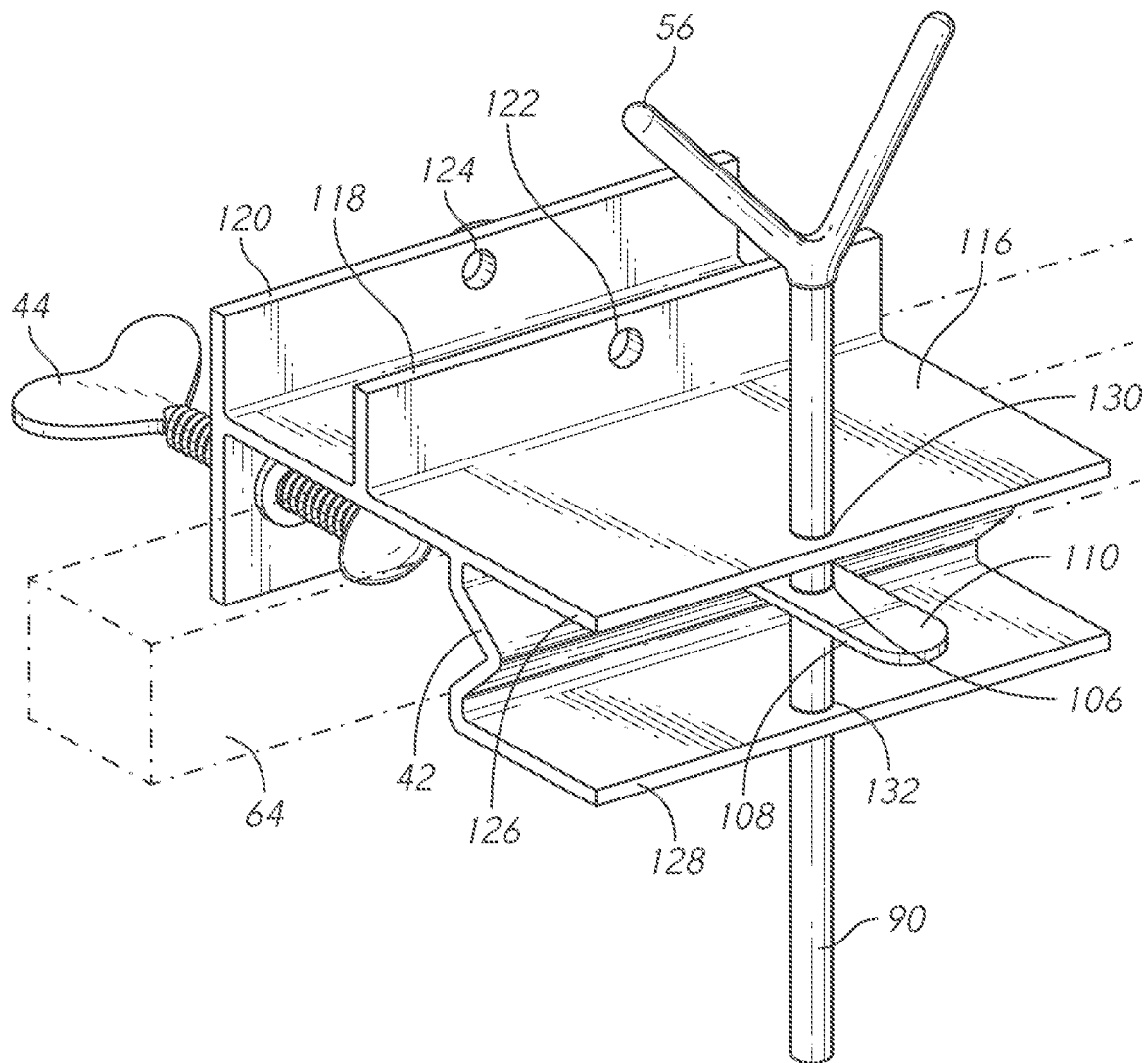


FIG. 15

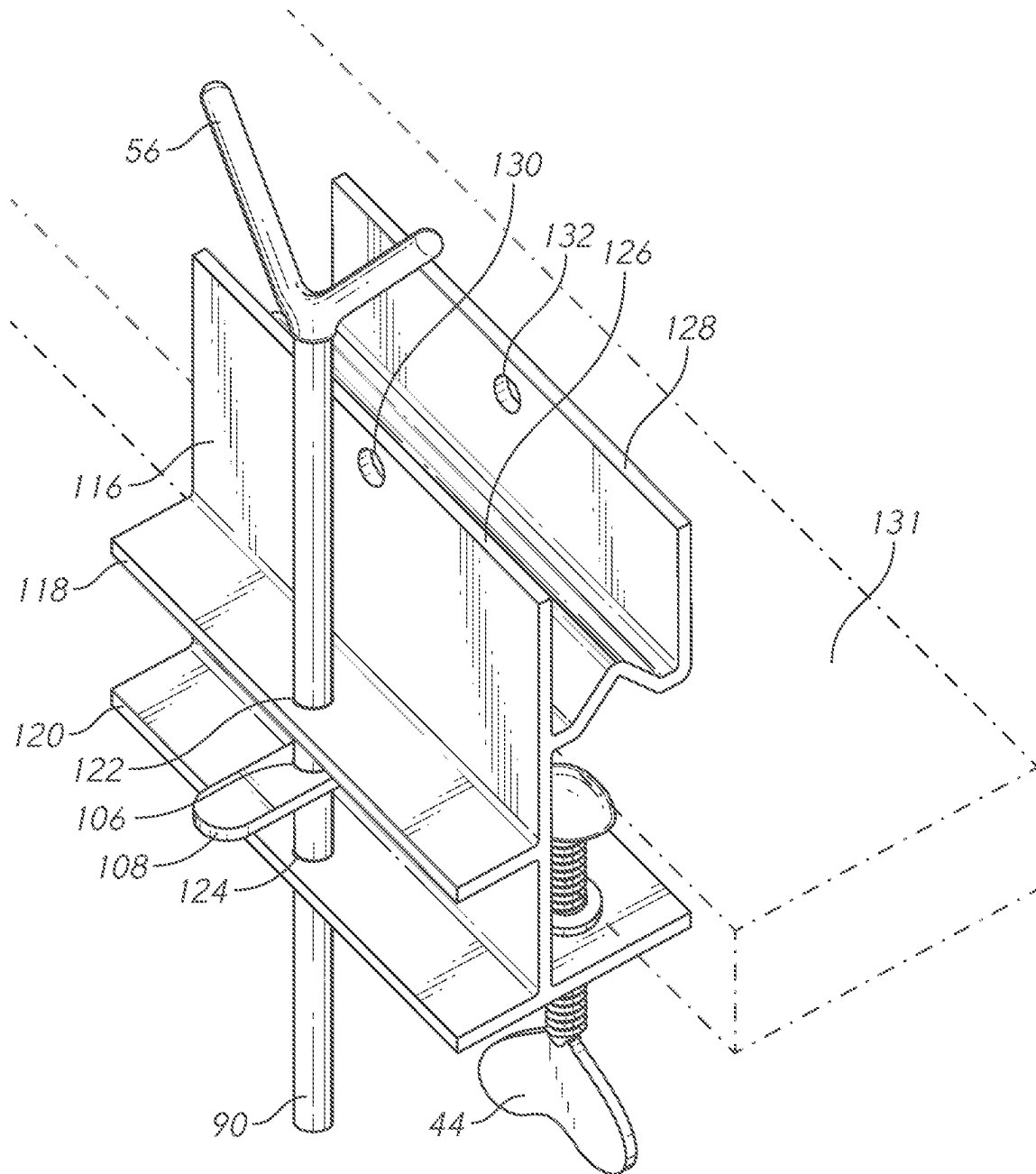


FIG. 16

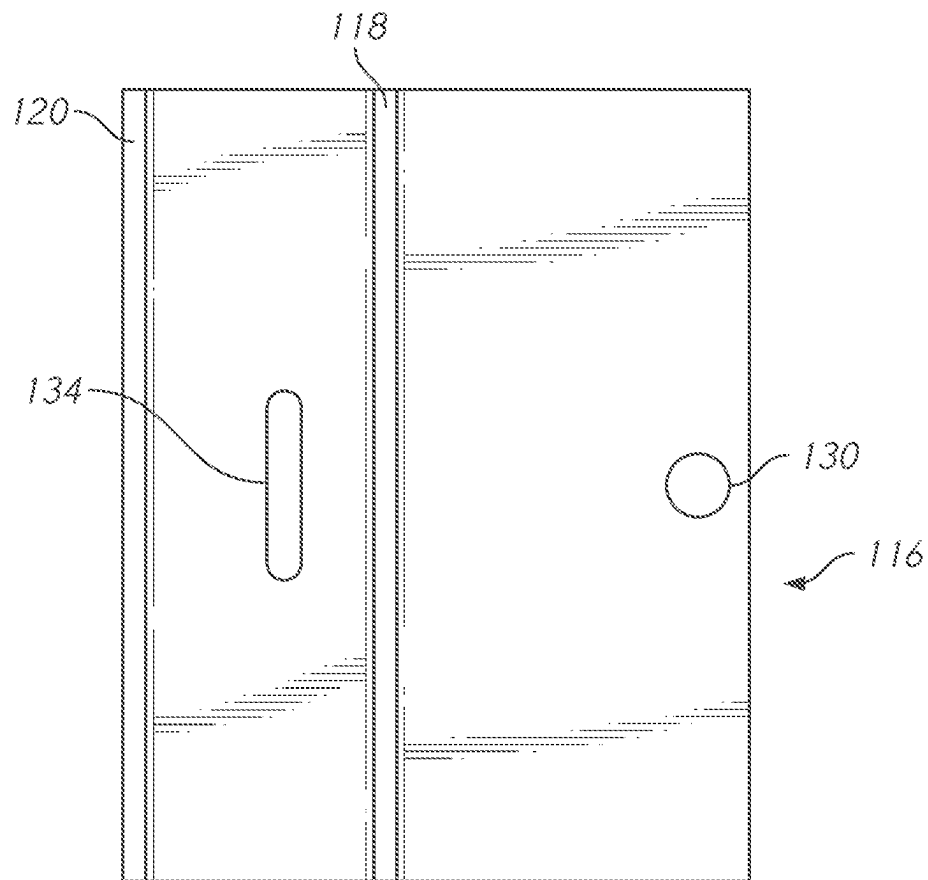


FIG. 17

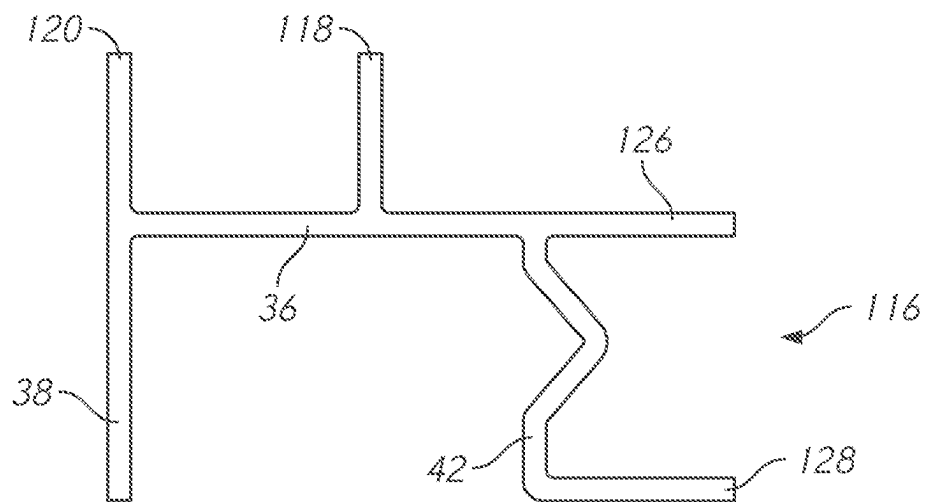


FIG. 18

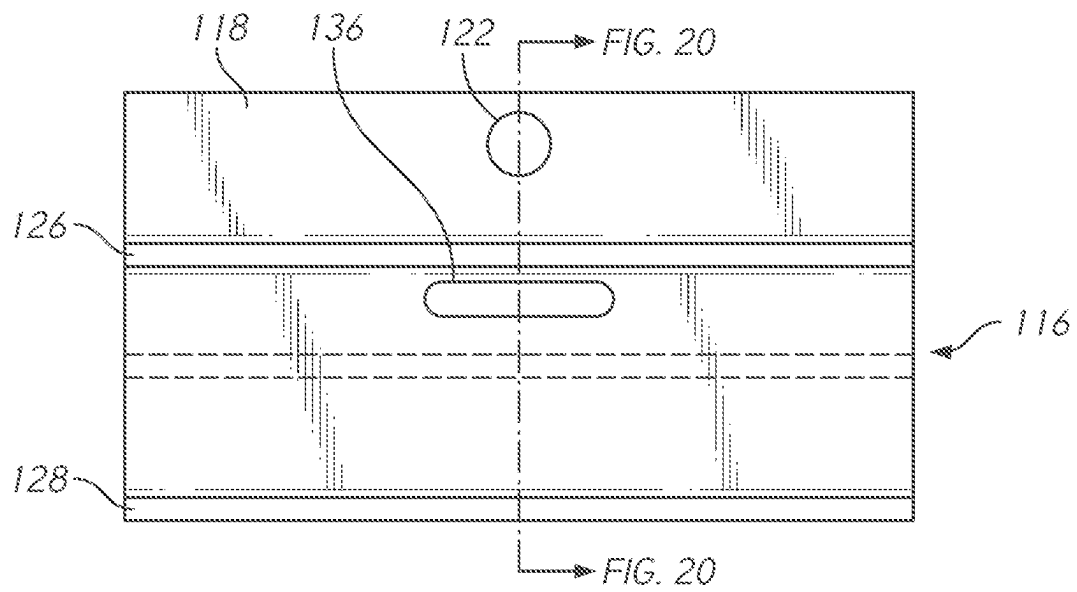


FIG. 19

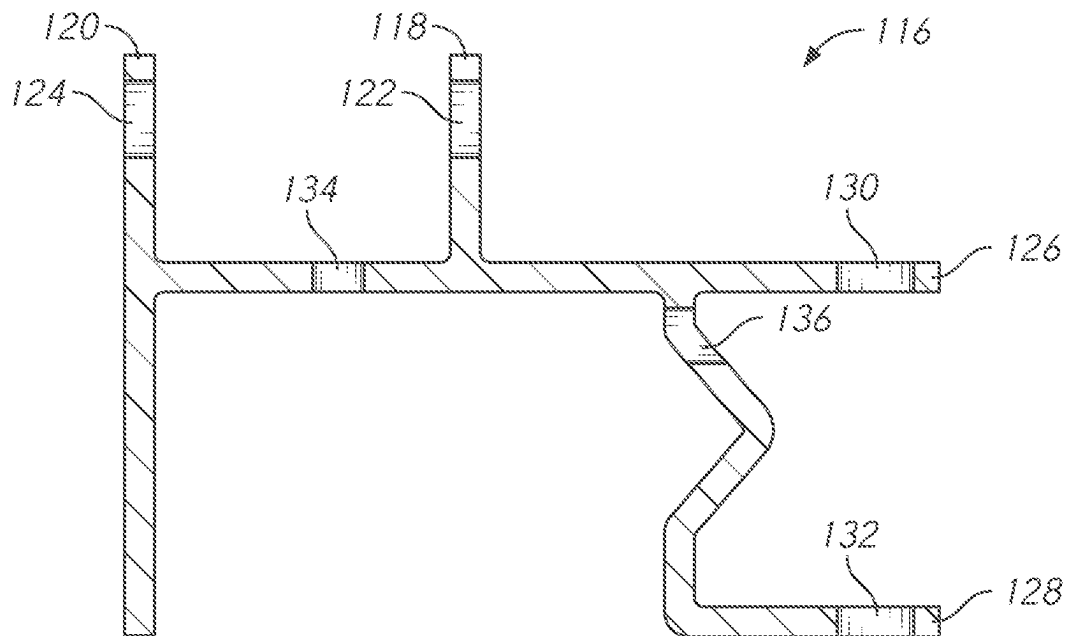


FIG. 20

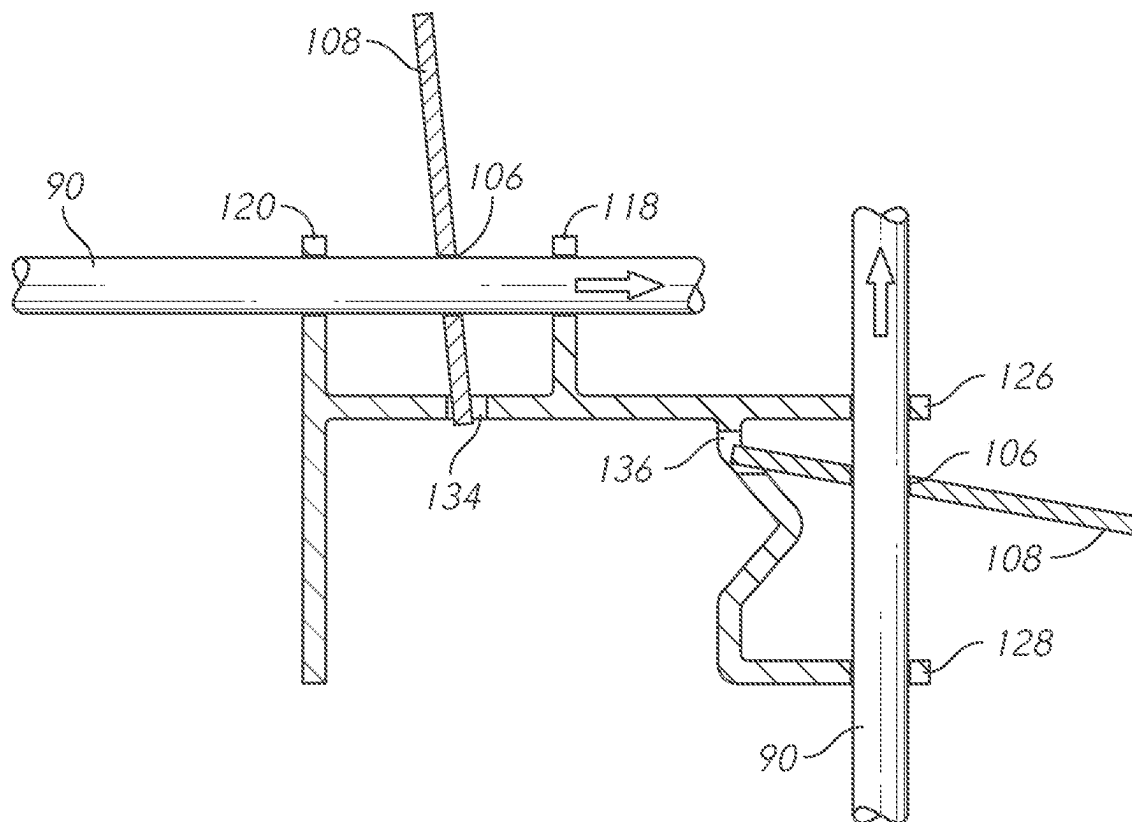


FIG. 21

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ADJUSTABLE FIREARM STEADY REST**CROSS-REFERENCES TO RELATED APPLICATIONS**

This application is a continuation-in-part of U.S. patent application Ser. No. 17/687,965. The parent application was filed Mar. 7, 2022. It listed the same inventor.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

Not Applicable

MICROFICHE APPENDIX

Not Applicable

BACKGROUND OF THE INVENTION**1. Field of the Invention**

This invention relates to the field of shooting. More specifically, the invention comprises an adjustable firearm support that can be attached to a variety of structures.

2. Description of the Related Art

Those familiar with the shooting sports know that the use of a steady rest greatly increases firearm accuracy. FIGS. 1 and 2 illustrate scenarios in which a steady rest would be beneficial. FIG. 1 illustrates a prior art tree stand. There are many different types of tree stand. Most, however, include a seat assembly 12, a foot plate 14, and a safety bar 16. The example shown in FIG. 1 includes a ladder. Hunter 18 climbs the ladder and sits in seat assembly 12. Safety bar is in a raised position when the hunter enters or exits the tree stand.

Once seated, the hunter pivots safety bar 16 down to the position shown. The safety bar is pivotally attached to the stand by a pair of pivot joints 15—only one of which is visible in FIG. 1. The safety bar is held in place by gravity. It actually provides a suitably stable rest point for shooting, but its position is too low. The hunter in FIG. 1 has assumed a typical shooting stance required to aim and fire rifle 20 from a tree stand. Trigger hand 22 is closed around the grip just to the rear of the trigger guard. Support hand 24 rests beneath the rifle's forend. This is in effect a modified "offhand" shooting position. It cannot be considered a seated shooting position because the hunter's elbows cannot be rested on any portion of the hunter's legs. Offhand is an unstabilized shooting position that significantly limits accuracy.

FIG. 2 shows a second common scenario. Shooting blind 26 conceals hunter 18. Opening 28 is provided for viewing and shooting. When a shot is to be taken, hunter 18 extends rifle 20 through opening 28. Sill 30 can provide a steady rest. A towel or sandbag is sometimes placed on the sill or on a shelf provided on the inside of the blind just below the sill. These devices do provide stability but are often cumbersome and inflexible to use. The result is that the hunter may wind up shooting off-hand as depicted in FIG. 2.

Many other shooting scenarios do of course exist, and the invention is by no means limited to the particular applications depicted. However, in general, it is desirable to provide

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a portable steady rest that can be clamped to a variety of structures and adjusted in height. The present invention provides such a device.

BRIEF SUMMARY OF THE INVENTION

The present invention comprises an adjustable shooting steady rest configured to clamp to a wide variety of structures. The steady rest includes a chassis with one or more clamps opposing a clamping bracket. The user places the chassis over a structure such as a safety bar or windowsill. The user then tightens the clamp(s) to lock the chassis in place. The chassis includes a rod receiver. A rod is adjustably attached to the rod receiver. A firearm support is provided on the upper portion of the rod. The firearm support is configured to support a suitable portion of a firearm—such as a rifle forend or barrel. The user adjusts the rod's position to provide a desired height for the firearm support.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

FIG. 1 is a perspective view, showing a hunter in a prior art tree stand.

FIG. 2 is a perspective view, showing a hunter in a prior art shooting blind.

FIG. 3 is a perspective view, showing a first embodiment of the inventive steady rest.

FIG. 4 is a side elevation view, showing the embodiment of FIG. 3 clamped to a large tube.

FIG. 5 is a side elevation view, showing the embodiment of FIG. 3 clamped to a small tube.

FIG. 6 is a side elevation view, showing the embodiment of FIG. 3 clamped to a square tube.

FIG. 7 is a side elevation view, showing the embodiment of FIG. 3 clamped to a sill.

FIG. 8 is a perspective view, showing the invention clamped to a safety bar on a tree stand.

FIG. 9 is a perspective view, showing the invention clamped to a sill in a shooting blind.

FIG. 10 is a sectional elevation view, showing an embodiment of the rod receiver using threaded inserts.

FIG. 11 is a sectional elevation view, showing an embodiment of the clamp mount leg using threaded inserts.

FIG. 12 is a sectional elevation view, showing an alternate embodiment for the adjustable attachment between the rod and the rod receiver.

FIG. 13 is a perspective view, showing an upward adjustment of the rod in the version of FIG. 12.

FIG. 14 is a perspective view, showing a downward adjustment of the rod in the version of FIG. 12.

FIG. 15 is a perspective view, showing an alternate embodiment of the chassis clamped in a horizontal orientation.

FIG. 16 is a perspective view, showing the embodiment of FIG. 15 clamped in a vertical orientation.

FIG. 17 is a top view showing the chassis of FIG. 15.

FIG. 18 is a side view showing the chassis of FIG. 15.

FIG. 19 is a front view showing the chassis of FIG. 15.

FIG. 20 is a side section view showing the chassis of FIG. 15.

FIG. 21 is a side section view showing how the chassis of FIG. 15 can be used in the horizontal orientation and the vertical orientation.

REFERENCE NUMERALS IN THE DRAWINGS

10 tree stand

12 seat assembly

14 foot plate
 15 pivot joint
 16 safety bar
 18 hunter
 20 rifle
 21 forend
 22 trigger hand
 24 support hand
 26 shooting blind
 28 opening
 30 sill
 32 steady rest
 34 chassis
 36 span leg
 38 clamp mount leg
 40 back leg
 42 clamping bracket
 43 first leg
 44 screw clamp
 45 second leg
 46 screw clamp
 48 insert
 50 insert
 52 threaded rod
 54 rod receiver
 56 firearm support
 58 foot
 60 large tube
 62 small tube
 64 square tube
 66 tip
 68 tip
 70 weld
 72 weld
 74 weld
 76 weld
 79 barrel
 80 insert
 82 insert
 84 female thread
 86 female thread
 88 tube
 90 smooth rod
 92 insert
 94 insert
 96 spring
 98 inner slot
 100 outer slot
 102 hole
 104 lock plate
 106 hole
 108 lock plate
 110 release
 112 fulcrum
 114 shelf
 116 unified chassis
 118 vertical mount upper rib
 120 vertical mount lower rib
 122 vertical mount upper hole
 124 vertical mount lower hole
 126 horizontal mount upper rib
 128 horizontal mount lower rib
 130 horizontal mount upper hole
 131 shelf
 132 horizontal mount lower hole
 134 vertical mount lock plate slot
 136 horizontal mount lock plate slot

DETAILED DESCRIPTION OF THE INVENTION

The present invention can be physically realized in a wide
 5 variety of embodiments. The following description provides
 details of a few of these. Many more embodiments will
 occur to those skilled in the art upon reviewing this disclo-
 sure. The scope of the invention should therefore be fixed by
 the claims rather than the descriptions of the disclosed
 10 embodiments.

In general, the invention includes (1) a chassis configured
 to clamp to a variety of external structures, and (2) an
 adjustable rod extending from the chassis, with a firearm
 support being provided on the rod. FIG. 3 shows a first
 15 embodiment of steady rest 32. Chassis 34 in this version is
 a section of C-channel. The three legs of the C-channel are
 clamp mount leg 38, span leg 36, and bracket leg 40.

Span leg 36 will ordinarily be placed above the external
 20 structure. Bracket leg 40 will normally be placed on the side
 of the external structure that is opposite to the position of the
 hunter. Clamp mount leg 38 will normally be placed on the
 same side of the external structure as the hunter. The reader
 should bear in mind, however, that directional terms such as
 25 “above” are arbitrary as steady rest 32 can be attached in a
 variety of orientations.

Clamp mount leg 38 in this example mounts a pair of
 screw clamps 44, 46. Threaded inserts 48, 50 are pressed
 into clamp mount leg 38. Each insert includes a threaded
 30 through-hole sized to accommodate the threads of a screw
 clamp. Clamping bracket 42 is attached to bracket leg 40. In
 operation, the screw clamps 44, 46 squeeze the external
 structure against clamping bracket 42—thereby attaching
 steady rest 32 to the external structure.

A central purpose of the steady rest is the provision of a
 35 stable firearm support 56. In the example of FIG. 3, firearm
 support 56 is provided on an end of threaded rod 52.
 Threaded rod 52 is adjustably attached to rod receiver 54.
 Rod receiver 54—in turn—is connected to chassis 34. The
 40 rod receiver includes an internal female thread that engages
 the male thread on threaded rod 52. The extension of
 threaded rod 52 above the position of chassis 34 is accom-
 plished by turning threaded rod 52. For the embodiment
 shown the threaded rod is given a right-hand thread. Turning
 45 the threaded rod in a clockwise direction moves the threaded
 rod downward and turning the threaded rod in an anti-
 clockwise direction moves the threaded rod upward.

Clamping bracket 42 allows the chassis to be clamped to
 a wide variety of external structures. FIGS. 4-7 provide
 50 examples of some of these external structures. In the
 example of FIG. 4, the invention is being clamped to large
 tube 60—the large tube being the external structure in this
 case. Large tube 60 is part of the safety bar on a tree stand.
 It has an external diameter of 1.250 inches (31.75 mm).

Clamp bracket 42 is fixedly attached to bracket leg 40 and
 span leg 36 by welds 72 and 70 respectively. These welds are
 preferably just small welds made at each end of the C-chan-
 nel and clamp bracket 42. The clamp bracket in this example
 is a simple angle piece having first leg 43 and second leg 45.
 60 The clamp bracket is rotated 45 degrees so that first leg 43
 is rotated 45 degrees with respect to bracket leg 40. The
 intersection of the two bracket legs 43, 45 lies on the central
 axis of screw clamp 44. When screw clamp 44 is tightened,
 foot 58 is pressed against large tube 60 and clamp bracket 42
 65 is pulled against large tube 60 as shown. The result is that
 three points of contact are established around the perimeter
 of the tube—thereby providing a solid engagement.

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FIG. 4 also shown how rod receiver 54 is joined to the C-channel in this embodiment. Two welds 74, 76 are made between the rod receiver and the C-channel. These are made at the point of tangency for rod receiver 54 (which in this example is a hollow tube).

FIG. 5 shows the same embodiment being clamped to small tube 62. Small tube 62 has an outside diameter of 0.875 inches (22.22 mm). Clamp bracket 42 accommodates the small tube. Tightening screw clamp 44 again propels foot 58 against the side of the tube-creating three points of contact and a solid engagement. The example shown works well for even smaller tubes-such as one having an outside diameter of only 0.500 inches (12.70 mm).

FIG. 6 shows the same embodiment being clamped to square tube 64. Square tube 64 has external dimensions of 1.000 inches by 1.000 inches (25.40 mm by 25.40 mm). Clamp bracket 42 accommodates this shape as well-again creating three points of solid contact around the perimeter of the external structure being clamped.

The examples of FIGS. 4-6 apply to tubes commonly used for the safety bar and other structures in tree stands. FIG. 7 pertains to structures such as are present in a blind like the one seen in FIG. 2. FIG. 7 shows the region of sill 30. Most blinds have some sort of sill across the bottom of each opening. These are often thicker than the tubes found in tree stands. A typical sill is 1.000 inches to 1.750 inches thick (25.40 mm to 44.45 mm). Span leg 36 is made long enough to accommodate a sill in the clamping gap. In the example shown, two tips 66, 68 of the legs of clamp bracket 42 bear against the outward facing side of sill 30. Foot 58 bears against the inward facing side of the sill when screw clamp 44 is tightened.

Many blinds include a shelf 114 just below sill 30. Such a shelf is useful for holding loose items like binoculars and ammunition boxes. However, such a shelf can also interfere with the operation of rod 52 if rod receiver 54 is placed on the side of the chassis that customarily resides inside the blind. For this reason, rod receiver 54 is placed on the side of the chassis that customarily resides outside the blind. The screw clamps are placed on the side that customarily resides inside the blind-so they can be more easily accessed by the hunter.

FIGS. 8 and 9 show the invention in use. In FIG. 9, steady rest 32 has been clamped to safety bar 16 of a tree stand. Threaded rod 52 has been adjusted vertically to place firearm support 56 at a desired height. Forend 21 of the rifle is placed on the firearm support and a steady shooting configuration is thereby created.

FIG. 9 shows the invention in use in a blind. Steady rest 32 has been clamped to sill 30. Threaded rod 52 has again been adjusted to a desired height for firearm support 56. In this example, rifle barrel 79 is actually placed in the support 56. A stable shooting configuration is again thereby created.

FIGS. 10 and 11 illustrate some construction details for the exemplary embodiment of FIG. 3. It is desirable to minimize the weight of steady rest. To that end, aluminum can be used for the chassis. Aluminum, however, may not provide adequate thread strength for the clamp screws. Inserts can be used to provide adequate thread strength and to minimize the need for potentially expensive threading operations.

FIG. 10 provides a sectional elevation view through the middle of rod receiver 54. In this version a simple hollow tube 88 provides the structure. Insert 80 is pressed into the open upper end of the tube. It includes female thread 84. Insert 82 is pressed into the open lower end of the tube. The two inserts include a female threaded passage 84. The length

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L of the hollow tube is set so that the threads in insert 82 align with the helical path of the threads in insert 80. This alignment allows a single threaded rod 52 to thread through insert 80 and insert 82.

In the preferred embodiments inserts 80, 82 are clinching steel inserts such as are manufactured by Penn Engineering and Materials, Inc., of Danboro, Pennsylvania, U.S.A.,. These inserts are assembled by providing (such as by punching or drilling) a properly sized hole and squeezing the insert into the hole (using fixturing on both sides). The insert plastically deforms the perimeter of the hole and becomes clinched to the base material.

FIG. 11 shows another similar insert that is pressed into clamp mount leg 38 in order to engage the threads of a screw clamp. Insert 48 has female thread 86 that is configured to match the thread of the screw clamps. Insert 48 is clinched to the material for clamp mount leg 38. The engagement is quite strong, so that the insert can withstand the forces generated by the screw clamp and transmit them to the chassis.

The example of FIGS. 3-11 used a threaded engagement to provide the adjustment between rod 52 and rod receiver 52. This is a very practical and robust adjustable engagement. Other adjustable engagements are possible, however. FIGS. 12-14 illustrate one of these other options.

The embodiment of FIGS. 12-14 uses a modified rod receiver. FIG. 13 is a perspective view of this embodiment. Rod receiver 54 is again attached to chassis 34—such as by welding. FIG. 12 is a sectional view through the middle of rod receiver 54. Smooth rod 90 has been substituted for threaded rod 52. Insert 92 is pressed into the top of tube 88 and insert 94 is pressed into the bottom. The inserts 92, 94 are preferably clinching steel inserts as described previously. However, each insert 92, 94 has a smooth bore through its center rather than a threaded bore. Smooth rod 90 is configured to smoothly slide through the two inserts. The bore through the two inserts 92, 94 in this example is 0.380 inches (9.652 mm). The diameter of the smooth rod in this example is 0.372 inches (9.449 mm). This combination allows guided but smooth motion of smooth rod 90 within inserts 92, 94.

A unidirectional latch is incorporated to hold rod 90 in position once a desired elevation has been set. The unilateral latch in this example is similar to the lock plate latch used in the drive mechanism of many caulking guns. Lock plate 108 is provided with a hole that is slightly larger than the diameter of smooth rod 90. Stacking lock plate 104 is provided with a hole 102 of the same diameter. In this example, the diameter of holes 102, 106 is 0.390 inches (9.906 mm). Inner slot 98 is provided in tube 88 proximate mounting leg 40. Outer slot 100 is provided in the opposite side of the tube.

The ends of the two lock plates 104, 108 nearest mounting leg 40 rest within inner slot 98. The lower portion of inner slot 98 becomes fulcrum 112. Compression spring 96 is sized to slide over smooth rod 90. This spring is compressed between the lower portion of insert 92 and lock plate 104. The spring tends to urge the two lock plates 104, 108 downward. The end of lock plate 108 within inner slot 98 is urged against fulcrum 112 and its further downward motion is thereby arrested. The two lock plates then pivot in an anticlockwise position until further pivoting is prevented by the edges of holes 102, 106 binding against smooth rod 90.

In studying the geometry of FIG. 12, those skilled in the mechanical art will readily appreciate the following: (1) If a user grasps smooth rod 90 and pulls it upward, the rod will slide smoothly upward through the unidirectional latch created by the two lock plates, (2) If a user attempts to push

smooth rod **90** downward, the two lock plates will immediately bind against the rod and prevent further downward motion. Thus, rod **90** can move up but cannot move down.

Of course, there will be times when the user wishes to move smooth rod **90** down. Release **110**—which is simply an extension of lock plate **108** is provided for this purpose. If the user urges release **110** upward, then the edges of holes **102**, **106** will disengage from their binding position against smooth rod and the smooth rod will be free to descend. Significantly, however, the user has to keep release **110** in a raised position in order for rod **90** to descend. As soon as the user lets go of release **110**, spring **96** will pivot the lock plates back to the position shown in FIG. **12** and any further downward motion of rod **90** will be prevented.

FIGS. **13** and **14** illustrate the practical application of this unidirectional latch. In FIG. **13**, the user wishes to raise the position of firearm support **56**. The user simply grabs the support—or some portion of smooth rod **90**—and pulls it upward. The support freely moves upward and remains in position when it is released. Thus, the user is able to incrementally adjust firearm support **56** upward until a desired position is reached.

FIG. **14** shows a lowering operation. In order to lower firearm support **56**, the user must pull up on release **110** as indicated by the arrow. Rod **90** is then free to move down. The speed of descent can be regulated by introducing a frictional element. As an example, insert **94** can include a polymer O-ring that bears against smooth rod **90** and slows its descent when the unidirectional latch is released. Returning to FIG. **12**, an exemplary O-ring **116** is provided within insert **94**. In this example, a DELRIN O-ring is used. This O-ring adds friction so that the rod and attached firearm support does not abruptly drop when release **110** is pulled upward. The reader will thereby appreciate that the user can rapidly and easily adjust the position of rifle support **56** using the unidirectional latch and the selective release **110**.

It is preferable to avoid a weldment for the chassis in higher-volume production. FIG. **15** shows an alternate embodiment—denoted as unified chassis **116**. Unified chassis **116** has a constant cross-section. It is preferably made as an extrusion. The details of the extrusion will be described briefly before turning to its functional advantages.

FIG. **18** shows a side elevation view of unified chassis **116**. The reader will note that it includes clamp mounting leg **38**, span leg **36**, and clamping bracket **42** as for the embodiment of FIG. **4**. The function of these elements is the same as for the embodiment of FIG. **4**. Unified chassis **116** contains additional features that allow the use of the unidirectional latch for the firearm support (as in the example of FIGS. **12-14**) without the need for a separate tube **54** to house it. These features also allow the chassis to be clamped and used in two different orientations—horizontal and vertical. Still looking at FIG. **18**—ribs **126**, **128** are used to mount the rod and firearm support when it is clamped in the orientation shown. If the chassis **116** is rotated 90 degrees counter-clockwise and clamped in a vertical orientation, then ribs **118**, **120** are used to mount the rod and firearm support. These options are explained in more detail in the following.

FIG. **15** shows the alternate embodiment when clamped in a horizontal orientation to square tubing **64**. In this context, “clamped in a horizontal orientation” means that screw clamps **44** exert clamping force in a horizontal orientation. The advancing screw clamps squeeze square tube **64** against clamping bracket **42**—which preferably includes a vee-notch shape as shown so that it can be stably clamped to a square or round shape.

Horizontal mount upper rib **126** includes a transverse horizontal mount upper hole **130**. Horizontal mount lower rib **128** includes an aligned horizontal mount lower hole **132**. These holes **130**, **132** slidably receive smooth rod **90**—which mounts firearm support **56** as for the prior versions. Smooth rod **90** also passes through hole **106** in lock plate **108**. The far end of lock plate **108** is secured within a slot in unified chassis **116** (The slot is not visible in FIG. **15** but, but horizontal mount lock plate slot **136** is shown in the elevation view of FIG. **19**).

A unidirectional latch is thereby created. The free end of lock plate **108** is urged downward by gravity (supplemented by a spring if desired). If the user pulls up on firearm support **56**, smooth rod **90** smoothly slides through hole **106** and any desired vertical extension for firearm support **56** can be easily set. Smooth rod **90** cannot move down, however, unless the user lifts up on the end of lock plate **108** (denoted as release **110**). A compression spring may be added between lock plate **108** and upper rib **126** to pivot the lock plate downward. However, experimentation has shown that the spring is not essential. If the lock plate is made of a dense material—such as steel—then gravity alone is sufficient to rotate the free end downward and lock hole **106** against smooth rod **90**.

In this embodiment one can also easily remove the rod/support and the lock plate. These may be removed by pulling smooth rod **90** upward until it is free of all the holes **130**, **106**, **132** and removed entirely from unified chassis **116**. This also allows lock plate **108** to be pulled to the right and free of horizontal mount lock plate slot **136** (shown in FIG. **19**). The rod/support can then be placed in holes **122**, **124** and the lock plate installed in a second corresponding slot.

FIG. **16** shows this configuration. The reader will note that the unified chassis has been placed in a vertical orientation—meaning that screw clamps **44** now exert clamping force in a vertical orientation. This is a good orientation for clamping to a horizontal object such as shelf **131**. The far end of lock plate **108** is placed within a second slot in the chassis (The slot is not shown in FIG. **16** but is shown as vertical mount lock plate slot **134** in FIG. **17**). Hole **106** in the lock plate is aligned with vertical mount upper hole **122** and vertical mount lower hole **124**. Smooth rod **90** is then passed downward through the three aligned holes **122**, **106**, **124** to again create a unidirectional locking mechanism.

In reviewing the orientations shown in FIGS. **15** and **16**, the reader will note how the features of unified chassis **116** allow the invention to be clamped in a wide variety of orientations while still providing a vertical adjustment range for smooth rod **90** and firearm support **56**.

FIG. **20** provides a section view through the middle of unified chassis **116** (note position of callouts in FIG. **19**). This section view allows the reader to easily perceive the location of the aligned holes and the two lock plate slots. Vertical mount upper rib **118** includes vertical mount upper hole **122**. Vertical mount lower rib **120** includes vertical mount lower hole **124**. Vertical mount lock plate slot **134** is placed nearby as shown. Horizontal mount upper rib **126** includes horizontal mount upper hole **130**. Horizontal mount lower rib **128** includes horizontal mount lock plate slot **136** as shown.

FIG. **21** shows the same section location with the lock plate and the smooth rod shown installed. FIG. **21** shows both orientations in use at the same time. This is shown for convenience. In reality—only one of the two orientations would be in use at a time. For the horizontal clamping configuration, smooth rod **90** is placed through the aligned

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holes in ribs **126**, **128**. The reader will note how the inward end of lock plate **108** rests within horizontal mount lock plate slot **136**, with smooth rod **90** passing through hole **106** in the lock plate. The rod may be pulled freely in the direction indicated by the arrow, but the lock plate will prevent motion in the opposite direction (gravity will cause the lock plate to rotate clockwise until it rests in the latched position shown).

For the vertical clamping configuration, the smooth rod **90** is installed in ribs **118**, **120**. In FIG. **21** the smooth rod is shown in the horizontal configuration but in actual use the assembly shown will be rotated approximately 90 degrees in the counterclockwise direction from the orientation shown in FIG. **21**. The reader will note how the inward end of lock plate **108** rests within vertical mount lock plate slot **134**. This again creates a unidirectional latch. Smooth rod **90** can be pulled freely in the direction shown by the arrow. However, it cannot move in the opposite direction since gravity will rotate lock plate **108** in the counterclockwise direction and cause hole **106** in lock plate to lock against smooth rod **90**.

Each sets of ribs (**118**, **120** or **126**, **128**)—along with the aligned holes mounted on each rib—therefore comprise a new type of rod receiver that performs the same function as the more complex rod receiver shown in FIGS. **12-14**. The use of the unified chassis reduces the number of parts required to perform the same functions.

The reader will thus perceive how the embodiment of FIGS. **15** and **16** allows the invention to be clamped in a horizontal orientation, a vertical orientation, and other orientations therebetween. In the version shown no stop is provided on the end of smooth rod **90**—so that the smooth rod can be pulled entirely free of unified chassis **116**. No stop is provided on lock plate **108** either. This allows the smooth rod and the lock plate to be removed and reinstalled in the other holes and slot in order to transition from the horizontal to the vertical configuration or vice versa. In other versions a stop and/or a spring can be provided. The stop can be made removable so that the smooth rod can still be transitioned between the vertical and horizontal orientations.

Many other features and combinations of features can be provided for the invention. Examples include:

1. Aluminum can be used for chassis **34** and steel for the adjustable rod extending from the rod receiver.
2. The weldment shown for the chassis can be a unified piece made by various processes—such as extrusion.
3. The holes in the lock plates can be slots instead.
4. Many different types of screw clamps can be substituted for the type shown. An entirely different clamping mechanism can also be used—such as a caulking gun drive mechanism.
5. The firearm support can be any desired shape instead of the simple “vee” shown.
6. The firearm support can be covered with a compliant material such as a natural or synthetic rubber.
7. Many alternate designs for clamp bracket **42** can be substituted.

The preceding descriptions contain significant detail regarding the novel aspects of the present invention. They should not be construed, however, as limiting the scope of the invention but rather as providing illustrations of the preferred embodiments of the invention. Many other embodiments will occur to those skilled in the art. Thus, the scope of the invention should be fixed by the following claims, rather than by the examples given.

Having described my invention, I claim:

1. An adjustable steady rest for a user aiming a firearm, comprising:

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- (a) a chassis, including a clamp mount and a clamping bracket;
- (b) a clamp attached to said clamp mount and configured to clamp against said clamping bracket;
- (c) said chassis including a rod receiver with a first hole and a second hole;
- (d) a rod passing through said first and second holes, said rod being translatable with respect to said rod receiver, an amount of extension of said rod from said rod receiver being selectively adjustable by said user;
- (e) a firearm support mounted on a distal end of said rod; and
- (f) wherein said adjustable engagement between said rod and said rod receiver comprises a unidirectional latch configured to allow said user to pull said rod freely in a first direction but not in a second direction opposite said first direction.

2. The adjustable steady rest as recited in claim 1, wherein said rod receiver comprises:

- (a) a first rib, with said first hole passing through said first rib; and
- (b) a second rib offset from said first rib, with said second hole passing through said second rib and said second hole being aligned with said first hole.

3. The adjustable steady rest as recited in claim 1, wherein said unidirectional latch comprises:

- (a) a lock plate having a first end, and a second end, and a lock plate hole;
- (b) said chassis having a lock plate slot; and
- (c) said first end of said lock plate resting within said lock plate slot and said rod passing through said lock plate hole.

4. The adjustable steady rest as recited in claim 2, further comprising:

- (a) a third rib having a third hole, said third rib being oriented transversely to said first rib;
- (b) a fourth rib offset from said third rib and having a fourth hole aligned with said third hole; and
- (c) said third and fourth holes being configured to receive said rod.

5. The adjustable steady rest as recited in claim 1, wherein said adjustable engagement between said rod and said rod receiver comprises:

- (a) said rod having a smooth, cylindrical exterior;
- (b) a lock plate, including a passage configured to bind against said cylindrical exterior of said rod when said lock plate is tilted;
- (c) a spring configured to bias said lock plate to said tilted position; and
- (d) a release configured to bias said lock plate away from said tilted position.

6. The adjustable steady rest as recited in claim 1, wherein said clamp is a clamping screw.

7. The adjustable steady rest as recited in claim 6, wherein said clamping bracket is aligned with an axis of rotation of said clamping screw.

8. The adjustable steady rest as recited in claim 7, wherein said clamping bracket comprises:

- (a) a first leg angularly offset 45 degrees in the anti-clockwise direction from said axis of rotation of said clamping screw; and
- (b) a second leg angularly offset 45 degrees in the clockwise direction from said axis of rotation of said clamping screw.